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Emeritus Professor - American University of Sharjah
AUS | الجامعة الأمريكية في الشارقة American University of Sharjah

Title:

The Journy of Applied Research:
The Need/Discovery – Innovation – Impact – Sustainable Knowledge Economy.
Sharing Successful Innovations.

October 06, 2026

MEET THE PRESENTER:

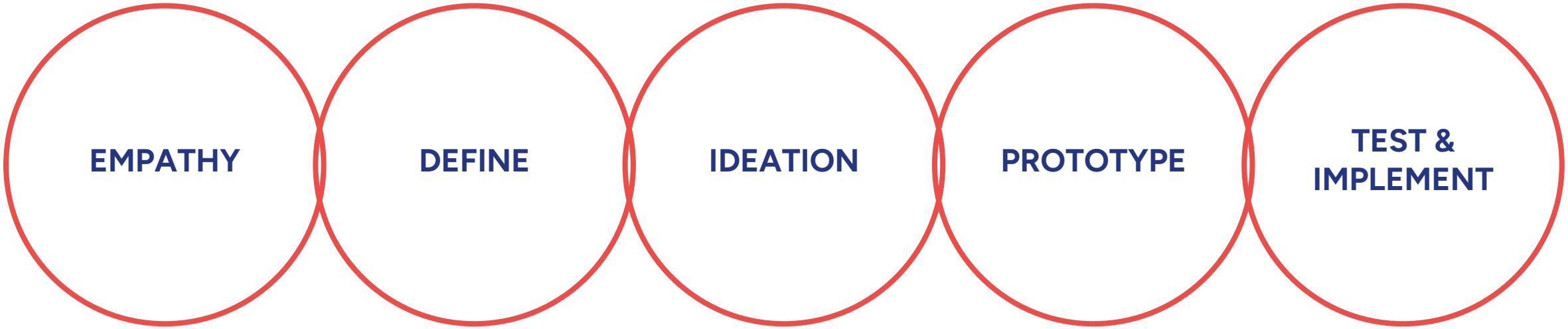
Adil K. Al Tamimi

Emeritus Professor – American University of Sharjah

وصف موجز للمساهمات المقدمة للمؤسسة :

- قدم البروفيسوعادل التميمي مساهمات استثنائية وتحولية للجامعة الأمريكية في الشارقة (AUS) منذ انضمامه إليها في عام 2000 كعضو هيئة تدريس مؤسس في كلية الهندسة .
- كانت قيادته ورؤيته والتزامه المستمر عوامل جوهرية في صياغة التطوير الأكاديمي والبحثي والمؤسسي للجامعة .
- هو صاحب براءة اختراع بالاشتراك مع أحد طلاب الماجستير .
-
- **مستشار وأكاديمي وباحث متميز يتمتع بخبرة تزيد عن 40 عاماً في مجالات التعليم العالي، والمناهج الدراسية، والاعتماد الدولي، والتصنيف، والبحث العلمي في كل من المملكة المتحدة، والولايات المتحدة الأمريكية، والعراق، ومنطقة الخليج العربي.**
- **تم تطوير وتدريس مقرر دراسي جديد مستوحى من جامعة ستانفورد بعنوان: آي إي إن 301 - عقلية الابتكار وريادة الأعمال.**
- **حصل على براءة الاختراع** رقم 8268 الممنوحة من وزارة الاقتصاد في دولة الإمارات العربية المتحدة، بموجب القرار الوزاري IP-2025-00491 بتاريخ 17 أبريل 2025 .
- تُعد هذه البراءة أول براءة اختراع إماراتية في مجال الخرسانة الإنشائية المطبوعة ثلاثياً، وتحمل عنوان: "تصميم خلطة خرسانية ذاتية التسليح للطباعة ثلاثية الأبعاد باستخدام مواد محلية من الإمارات ."
- لدي اختراعان جديدان تم تقديمهما في فبراير 2026
- حصل على الجائزة الأولى في دولة الإمارات العربية المتحدة عن استدامة المدن الجديدة في يونيو 2026.
- **صُنّف ضمن أفضل 1% عالمياً بناءً على المنشورات البحثية لعام 2024**، والذي أُعلن عنه في 25 يناير 2026 .
- الرابط https://x.com/aus_cen/status/2016150408411140104
-
- **هو عضو في مجمع محمد بن راشد للعلماء (MBRAS) في دولة الإمارات .**
- خبير مسجل في "خارطة الإمارات للبحث العلمي"، وهي مبادرة مشتركة بقيادة مجلس أبحاث التكنولوجيا المتطورة (ATRC)، وبالتعاون مع مجلس الإمارات للبحث والتطوير(ERDC) ، ووزارة التعليم العالي والبحث العلمي (MOHESR).
- الملف الشخصي للاقتباسات العلمية :(Google Scholar)
- الرابط https://scholar.google.com/citations?user=huaWm3UAAAAJ&hl=en :
- يبلغ عدد منشوراته العلمية 128 بحثاً في مجلات "سكوبس (Scopus) "والمؤتمرات الدولية المحكمة ذات معدلات الاقتباس العالية .

IT IS ALL ABOUT DESIGN THINKING?

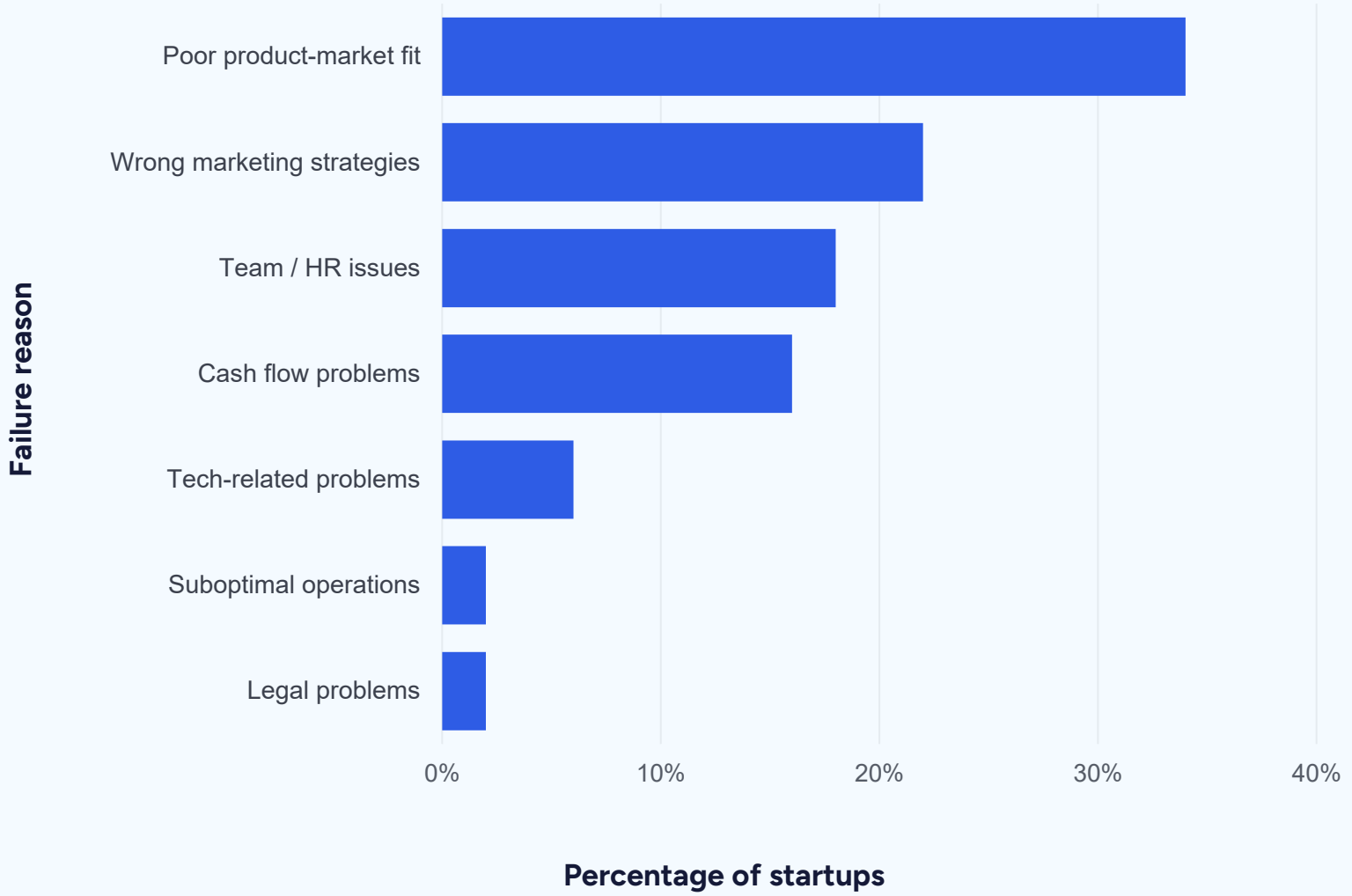


90% STARTUPS FAIL



https://www.forbes.com/sites/neilpatel/2015/01/16/90-of-startups-will-fail-heres-what-you-need-to-know-about-the-10/

Why Do Most Startups Fail?



https://explodingtopics.com/blog/startup-failure-stats

BUILD SOLUTIONS

THAT PEOPLE NEED

BUILD SOLUTIONS

THAT PEOPLE NEED



PAINKILLER



VITAMINS

THE HIPPO ROLLER

- Pettie Petzer and Johan Jonker were inspired by the **impact of water crisis in a rural environment.**
- They spoke with the locals, **understood the problem.**
- Found **other ways** that can help solve this issue.
- **Prototyped** with basic models
- **Learnt from feedback** and produced the final version.



THE HIPPO ROLLER









Source: The Hippo Roller

THE HIPPO ROLLER



Source:
Insider Tech

THE HIPPO ROLLER

- Pettie Petzer and Johan Jonker were inspired by the **impact of water crisis in a rural environment.**
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EMPATHY

DEFINE

IDEATION

PROTOTYPE

TEST & IMPLEMENT



THE JOURNEY OF A PRODUCT

PROBLEM

DEFINE

PROTOYPE

TEST

DESIGN

BUILD

Derived from: Waterfall Model

WHAT IS THE USE/ PURPOSE OF A USER JOURNEY MAP?

User Journey maps helps **visualize and document** a process/ experience a user has when they use a product/ service.

Helps uncovering hidden issues



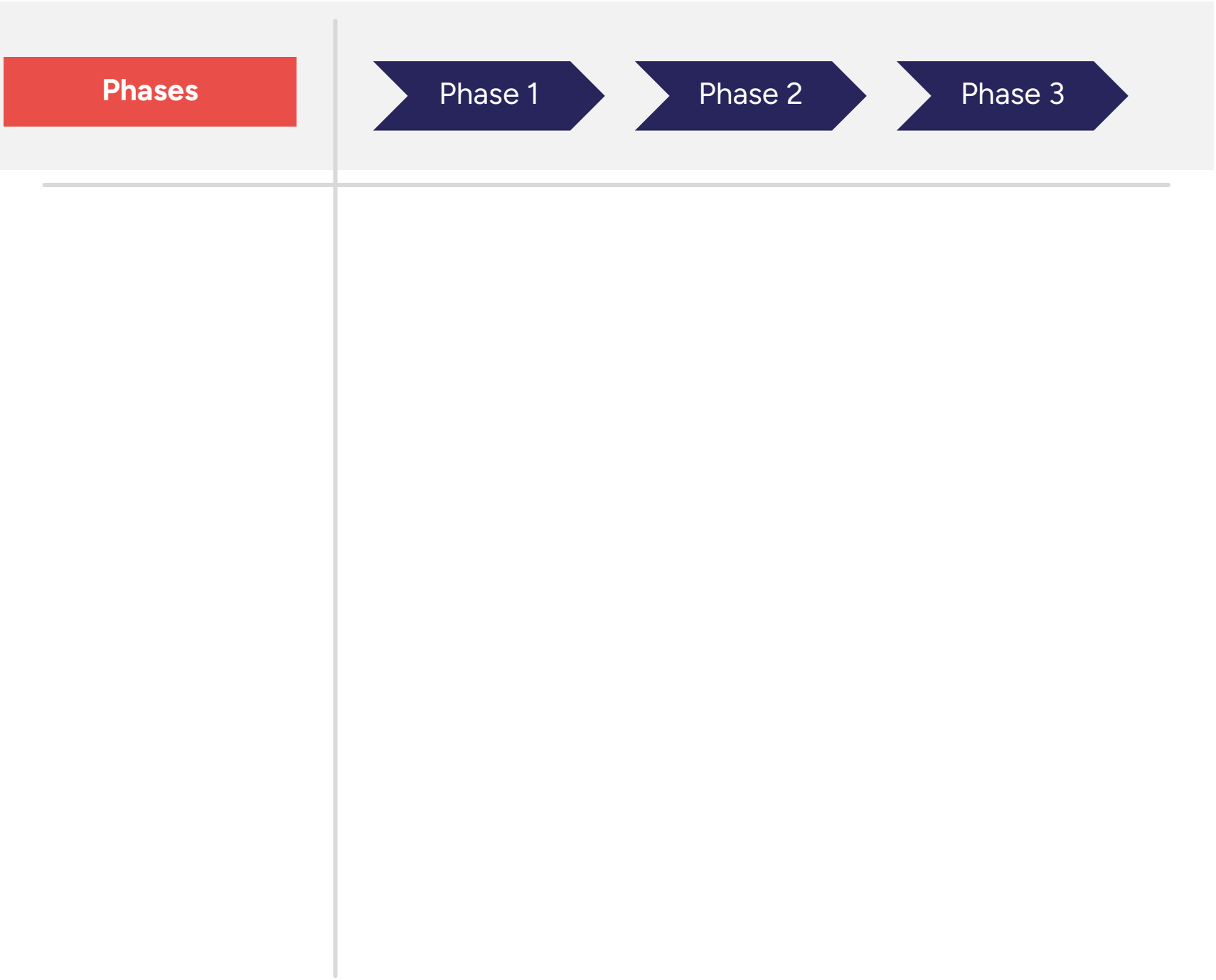
BEFORE YOU START...

DO YOUR RESEARCH!

- **Primary Research:** Face to face interviews, surveys
- **Secondary Research:** Google Trends, Web Analytics, Social Media Analysis

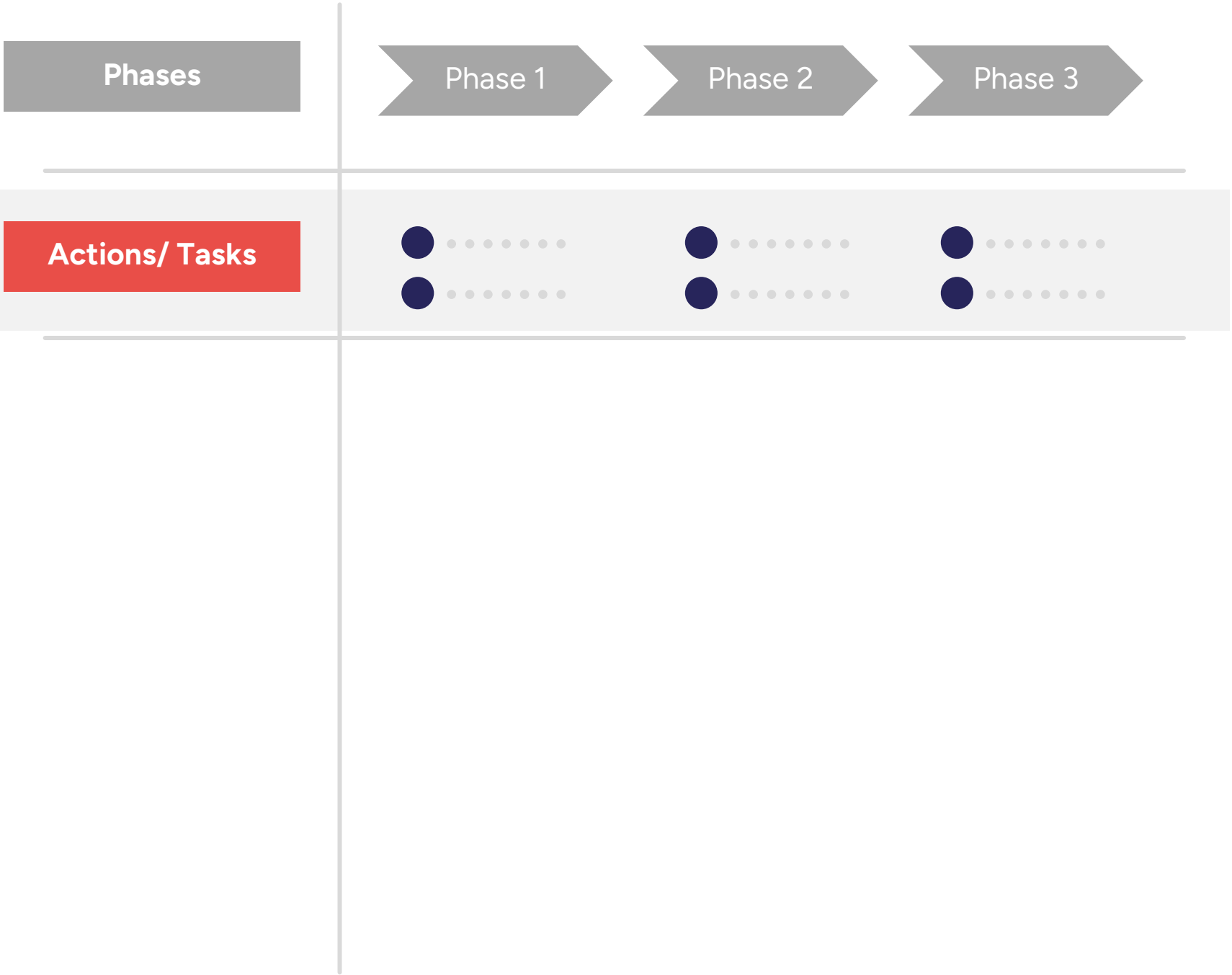
CREATE PERSONAS

- Identify your persona for the user journey map.
- What is their goals, characteristics?
- **Empathize!**



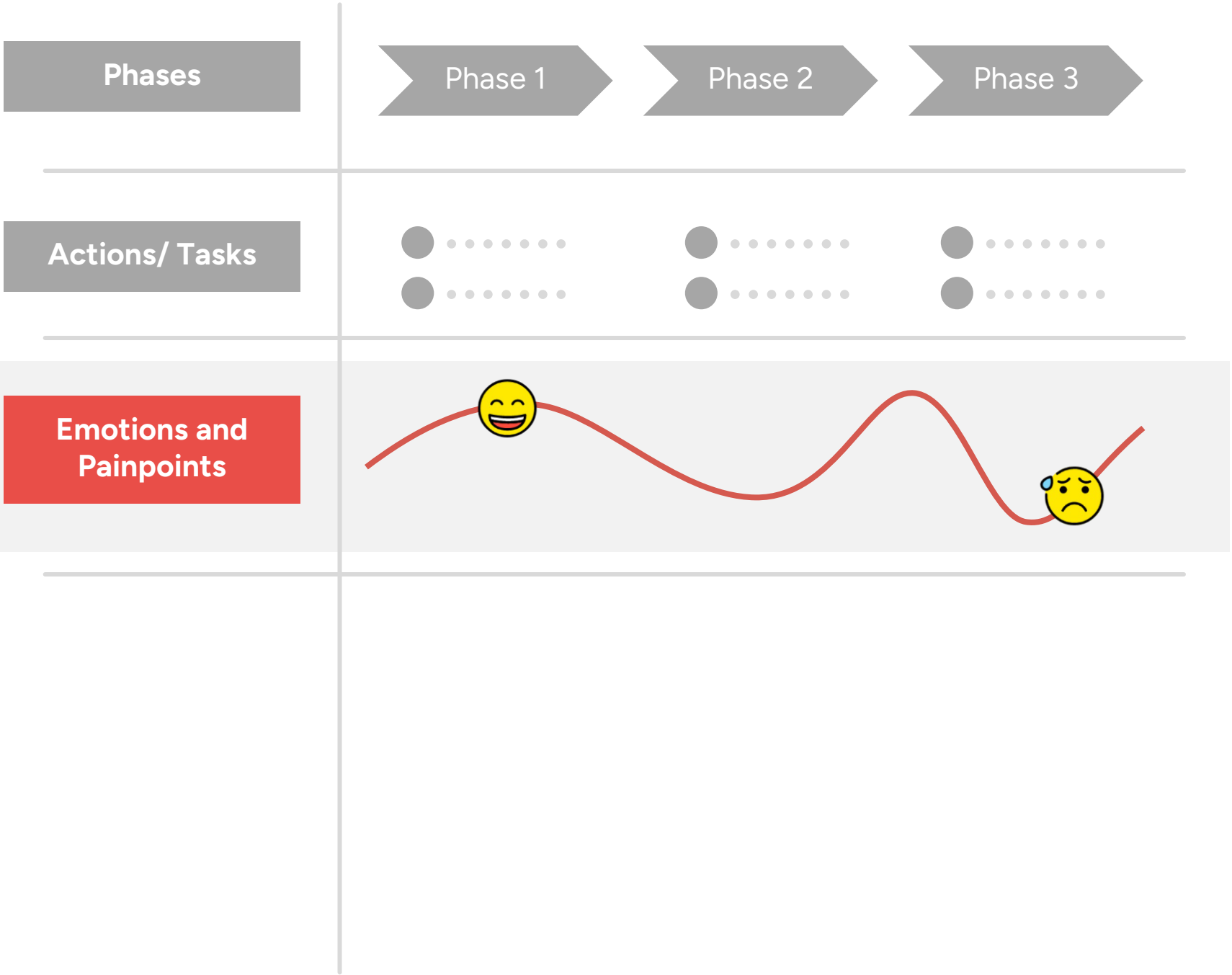
1. Phases

- Identify the phases of the process a user goes through.
- Write them down in chronological order.
- Note: Don’t forget what happens before and after they use product/ service.



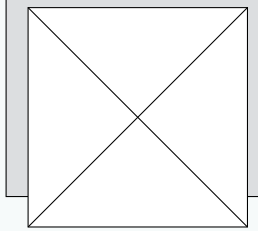
2. Actions/ Tasks

- Write down the detailed actions and tasks the user followed to reach their goal in each phase.



3. Emotions and Painpoints

- Map the users emotion across each phase.
- What are the positive emotions?
- **What are their Painpoints? What annoys, bothers them?**



SCENARIO

NAME

ABOUT USER/ PERSONAS

GOALS

CHARACTERISTICS

PHASES OF THE JOURNEY

ACTIONS/ TASKS







PAINPOINTS

TOUCHPOINTS AND CHANNELS

OPPORTUNITIES

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Personas



SCENARIO

Airport Experience (Pre-Covid)

NAME

Chantel Charles

ABOUT USER/ PERSONAS

- Mom of two kids (4 and 2 years old).
- Takes 3-4 family trips every year.
- Works at a tech firm and leads a very busy schedule.
- Is very tech savy.

GOALS

- Looking for a comfortable experience
- Looking for a peace of mind/ stress-free journey

CHARACTERISTICS

- Very busy schedule
- Loves easy and fast booking
- Highly organized
- Loves to Plan

PHASES OF THE JOURNEY

ACTIONS/ TASKS

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TOUCHPOINTS AND CHANNELS

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PHASES OF THE JOURNEY	Pre Flight (Anticipation)	Home to Airport	Check-in Counter	Immigration/ Security Check	In the Airport (Duty Free/ Lounge/ Food Court)	Boarding Gate (Waiting)	Boarding Experience
ACTIONS/ TASKS							
PAINPOINTS							
TOUCHPOINTS AND CHANNELS							
OPPORTUNITIES							

Phases

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PAINPOINTS							
TOUCHPOINTS AND CHANNELS							
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Actions

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PAINPOINTS							
TOUCHPOINTS AND CHANNELS							
OPPORTUNITIES							

Emotions

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PAINPOINTS		Carrying luggage around	Long wait time	Finding way around	Convenience for duty free/ food		
TOUCHPOINTS AND CHANNELS							
OPPORTUNITIES							

Painpoints

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PAINPOINTS	Find out information (App/ Web)	Carrying luggage around	Long wait time	Finding way around	Convenience for duty free/ food		
TOUCHPOINTS AND CHANNELS	Ensure all documents are valid (Check)	Book a taxi/ luggage drop off service (App/ Web)	Follow Airport signage	Interaction with Check-in agent	Interaction with Immigration Officer/ Security Check	Follow singage Check flight status/ boarding status	Interaction with boarding gate agent
OPPORTUNITIES							

Touchpoints

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PHASES OF THE JOURNEY

Pre Flight (Anticipation) | Home to Airport | Check-in Counter | Immigration/ Security Check | In the Airport (Duty Free/ Lounge/ Food Court) | Boarding Gate (Waiting) | Boarding Experience

ACTIONS/ TASKS

- Learn more about destination
- Ensure they have all documents ready
- Book a taxi to get to airport
- Carry luggage
- See if there is online check-in counter
- Find the correct check-in counter
- Wait in the queue
- Connect on airport Wi-Fi
- Find way to lounge/ food court/ duty free
- Shop/ Eat/ Rest/ Work (if any)
- Find the correct gate
- Keep a check on the boarding info
- Wait in the queue
- Board the flight

PAINPOINTS

- Carrying luggage around
- Long wait time
- Finding way around
- Convenience for duty free/ food

TOUCHPOINTS AND CHANNELS

- Find out information (App/ Web)
- Ensure all documents are valid (Check)
- Book a taxi/ luggage drop off service (App/ Web)
- Follow Airport signage
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- Interaction with Immigration Officer/ Security Check
- Follow singage
- Check flight status/ boarding status
- Interaction with boarding gate agent

OPPORTUNITIES

- Single Hub for all your travel needs?
- Check-in from home?
- Fast-track services?
- Easier way finding?
- Collect and go? Shop from home and collect at airport?

Opportunities

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Our Innovations on the 3D Concrete Printing

1. Oulu University – Finland 2017.
2. First Senior Design Project on 3D Concrete 2018
3. First 3D Concrete Traditional Villa in UAE – 2019
4. Four Senior Design Projects 2019-2023

5. Ten MSc. Theses and Four PhDs. 2020 - 2027:

- 5.1 MSc. 3D concrete mix design 2020 – 2021 - Graduated
- 5.2. MSc. 3D concrete structural design 2021 – 2022 - Graduated
- 5.3. MSc. 3D concrete durability assessment 2021- near Completion
- 5.4. MSc. 3D concrete safety – 2022 – ongoing
- 5.5. MSc. 3D concrete cost – ongoing
- 5.6. MSc. 3D concrete sustainability – ongoing
- 5.7. MSc. 3D concrete wall performance - Started
- 5.8. MSc. 3D concrete barrier in construction - Started
- 5.9. MSc. 3D concrete structural integrity - Just started
- 5.10 MSc. 3D concrete – slabs, column or foundation – Just Started

We took an active part on the construction of the first traditional 3d concrete villa constructed in Sharjah 2019



How it worked?

1. Introduction
2. Methodology
3. Experimental Program
4. Trial Batches
5. Fresh Properties
6. Shape Stability
7. Mechanical Properties
8. Conclusions

Introduction

Background → Motivation → Significance → Objectives

- **Improvement in Concrete Material:** Properties &



Manual mixing of concrete



Concrete Mixer



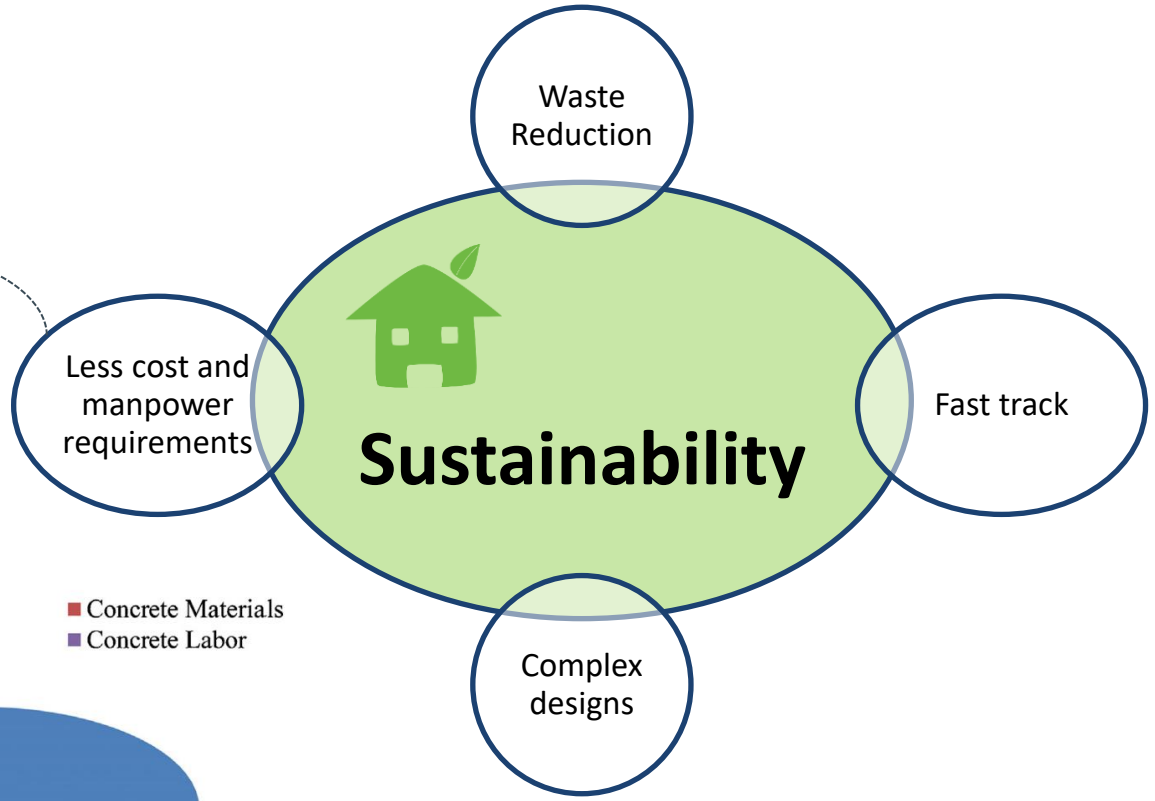
Concrete ready-mix truck



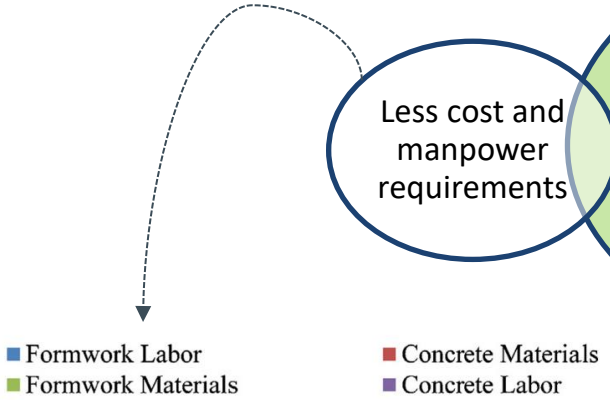
Lifting of a precast slab

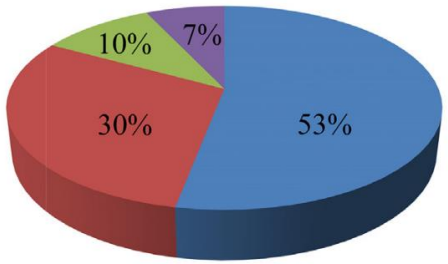
introduction

Background → Motivation → Significance → Objectives





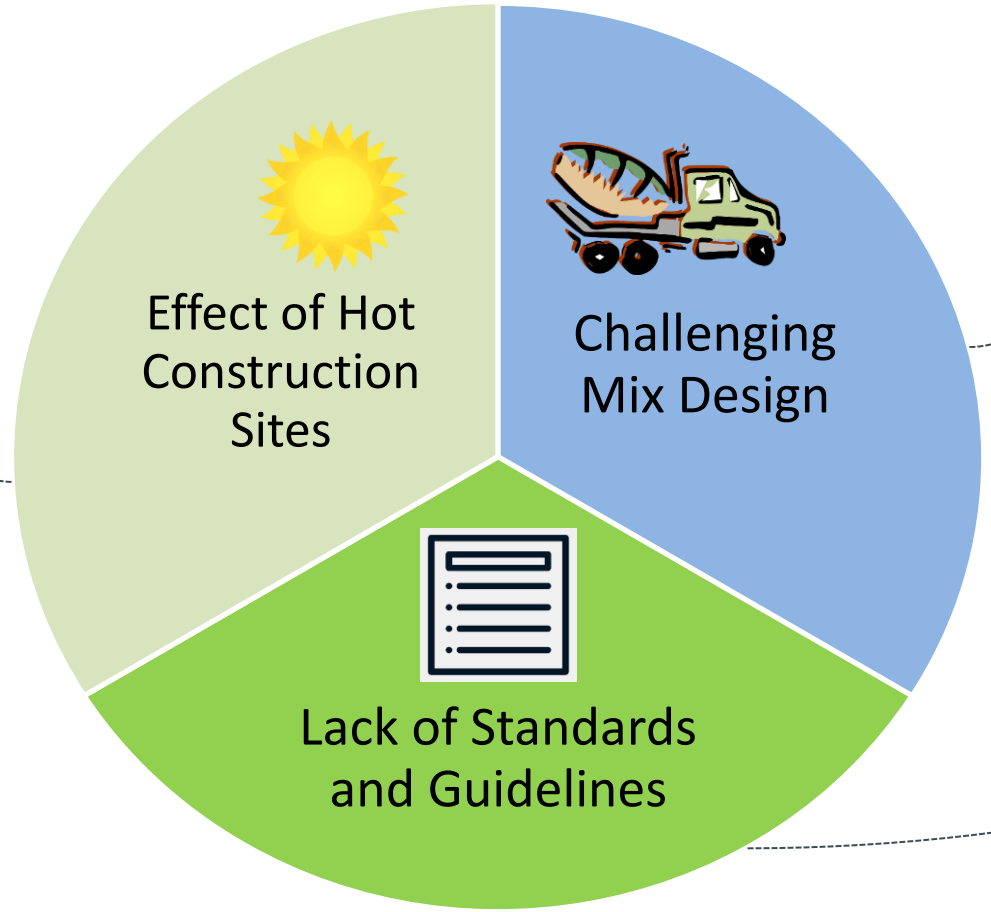




Cost Distribution for Concrete Works [19]

introduction

Background → Motivation → Significance → Objectives

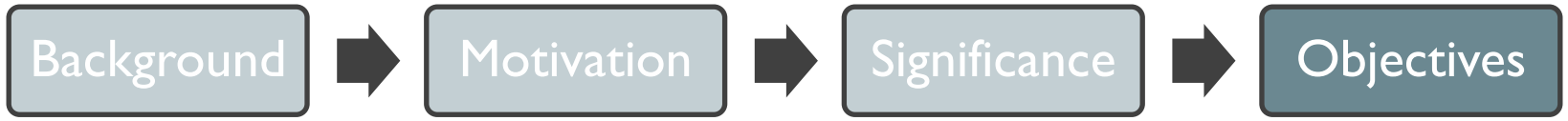


- UAE construction sites are hot most of the time in a year.
- Controlling environment at sites for printing will lead to large energy consumption, and this enhances the sustainable goal.

- The combination of flowability and quick hardening is tricky.

- Comprehensive criteria for performance evaluation is yet not available.

Objectives



- Design a concrete mixture that is suitable for the 3D printing applications.
- Evaluate rheological and hardened state properties of the developed mixture.
- Investigate bond shear strength of the interface between layers.
- Study the effects of the UAE's harsh environmental conditions on the material properties.



UAE Achievements



Office of the Future, Dubai [13]



Sharjah 3D House [14]



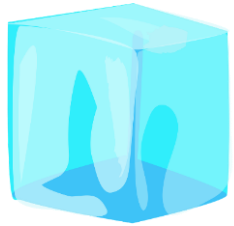
Warsan Buidling, Dubai [15]

Methodology

Experimental Scenarios

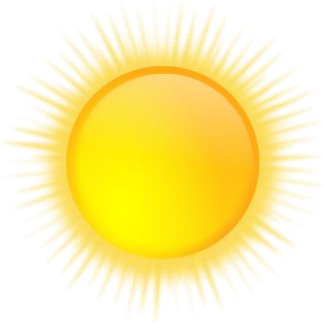
Scenario 1:

- Ambient conditions.
- Storage in lab.



Scenario 2:

- Site conditions.
- Storage outside the lab.





Methodology

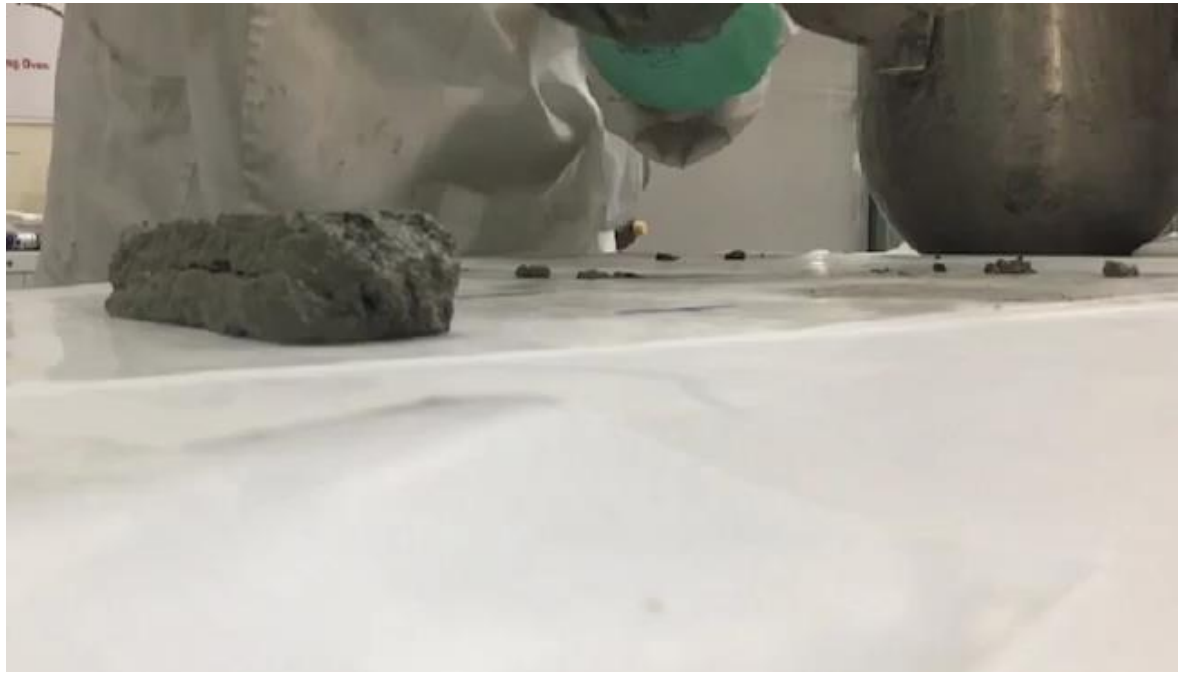
Trial Batches and Extrusion

Hollow-core Concrete Slab Mix Design [34]

Ingredient	Corrected Batch Weight (kg/m^3)
Cement	288
GGBS	162
Water	130
10 mm Aggregate	1073
0-5 mm Aggregate	558
Dune Sand	224
Water Reducing Admixture	4

Methodology

Trial Batches and Extrusion



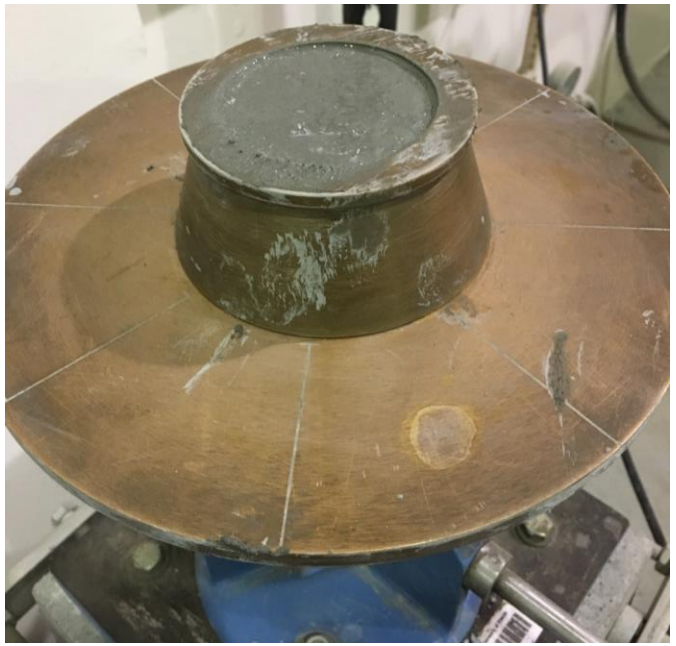




Methodology

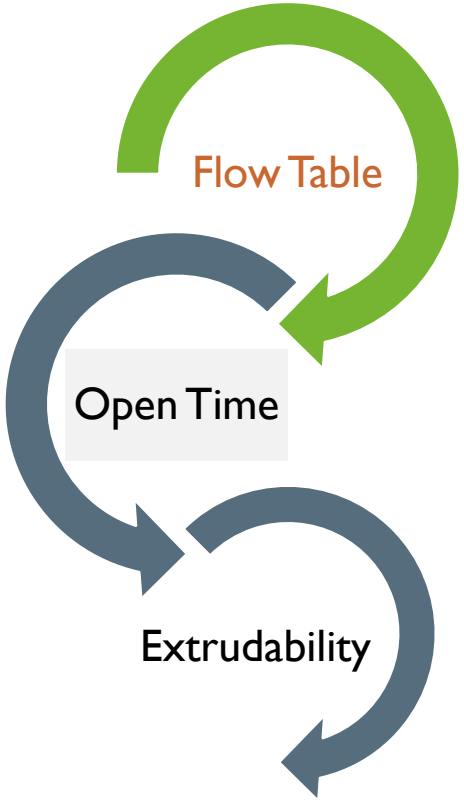
Fresh Properties

- ASTM C230.
- % Flow = $\frac{Increase\ in\ flow\ diameter}{Original\ diameter} * 100\%$ [35].





Flow Table Test





Methodology

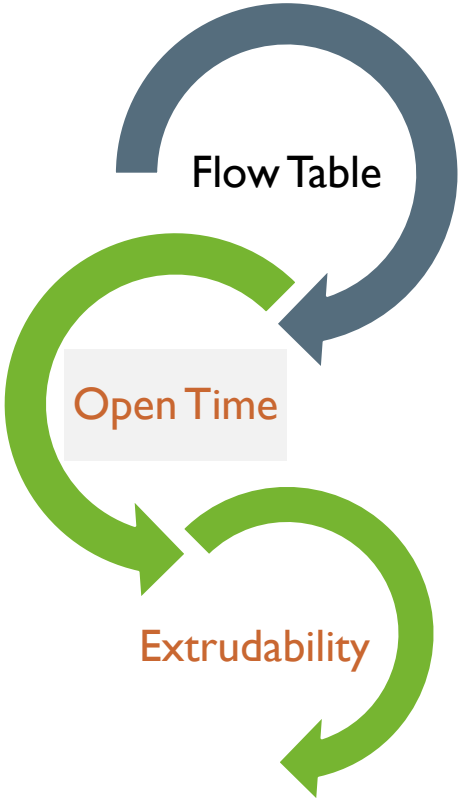
Fresh Properties

Open Time:

- The time when mix is ready until printed layers start to show stoppage, blockage, or discontinuities.

Extrudability:

- Subjective Assessment.
- The ability of a mix to be extruded smoothly without stoppage, while shape is preserved with good quality.





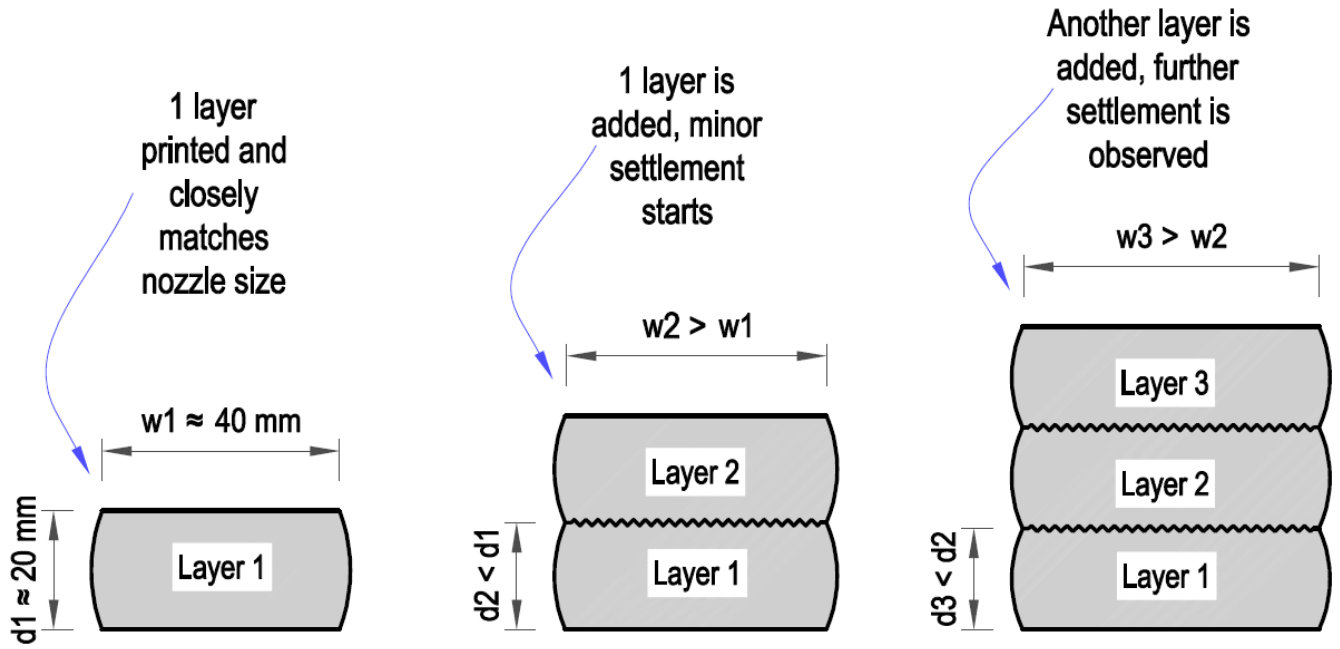
Methodology

Shape Retention

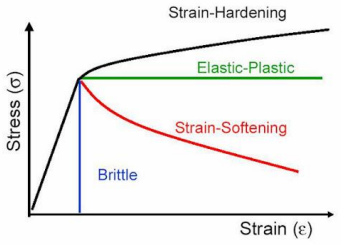
- Flexural and bond test specimens are used.
- Shape Retention Index (SRI).
- Height Retention Index (HRI).

- SRI = $\frac{Nozzle\ Width\ (40mm)}{Actual\ Print\ Width}$

- HRI = $\frac{Total\ Height\ of\ Printed\ Part}{Optimal\ Height}$



Settlement of Concrete Layers while Additional Layers are Added.



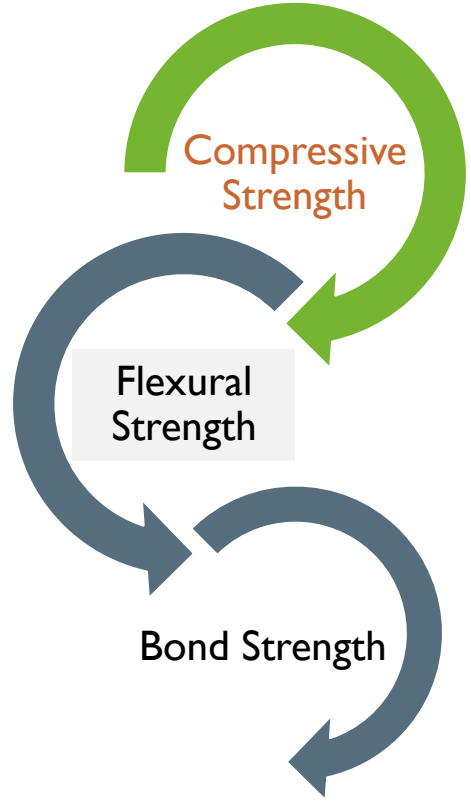
Methodology

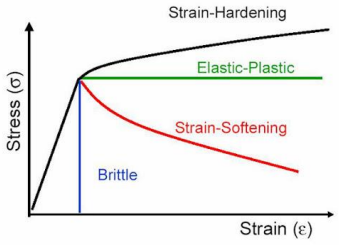
Mechanical Properties

- ASTM C109.
- Compressive Strength = $\frac{Failure\ Force\ (N)}{Area\ under\ Pressure\ (mm^2)}$ [36].



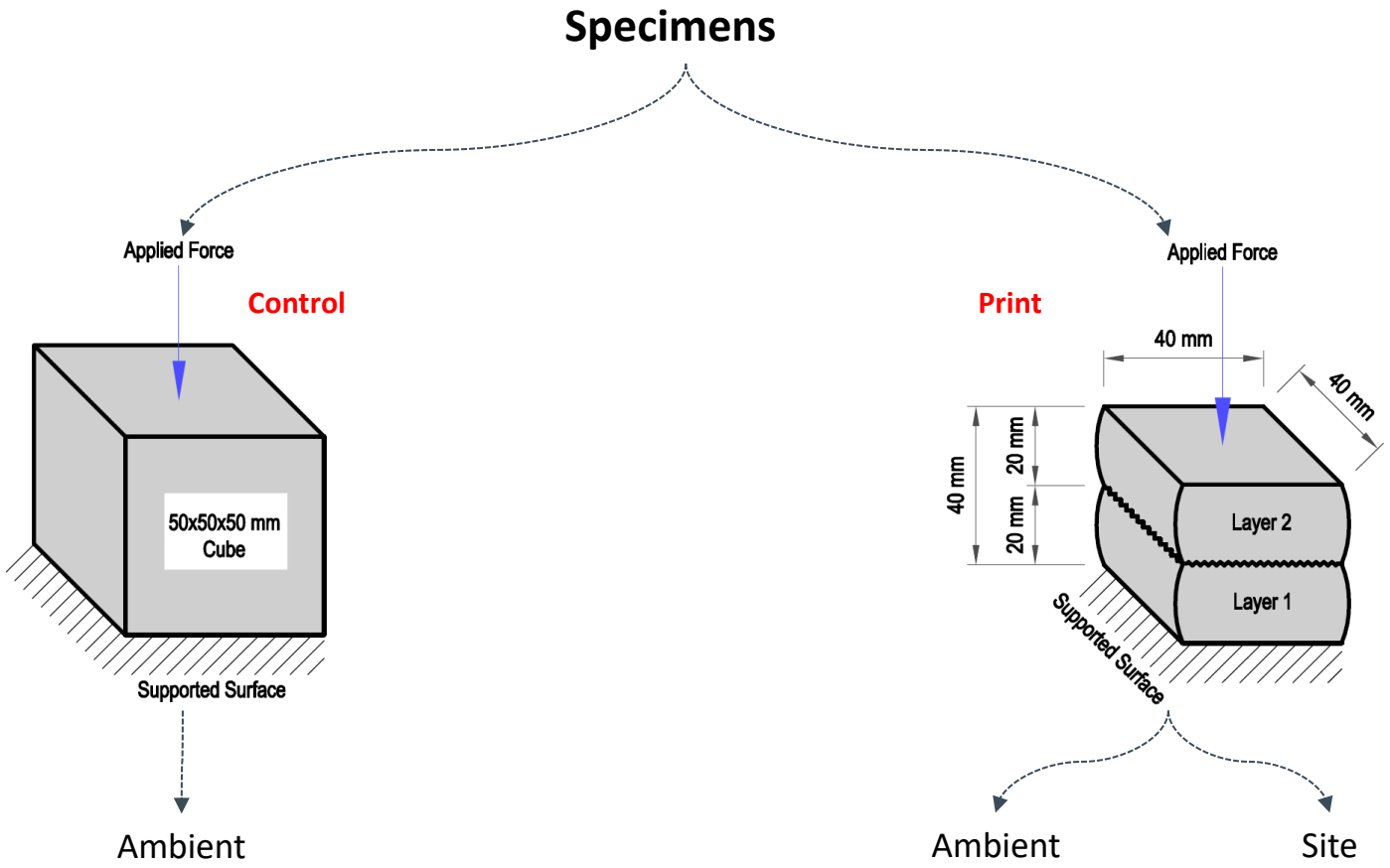
Compression Test Setup [37]

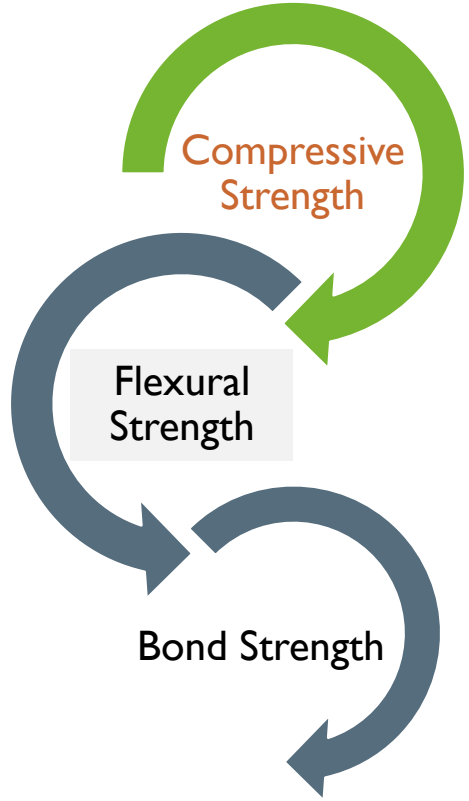


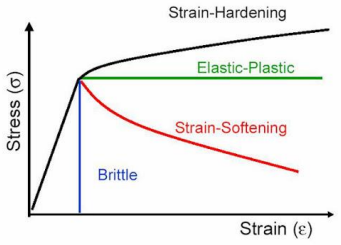


Methodology

Mechanical Properties







Methodology

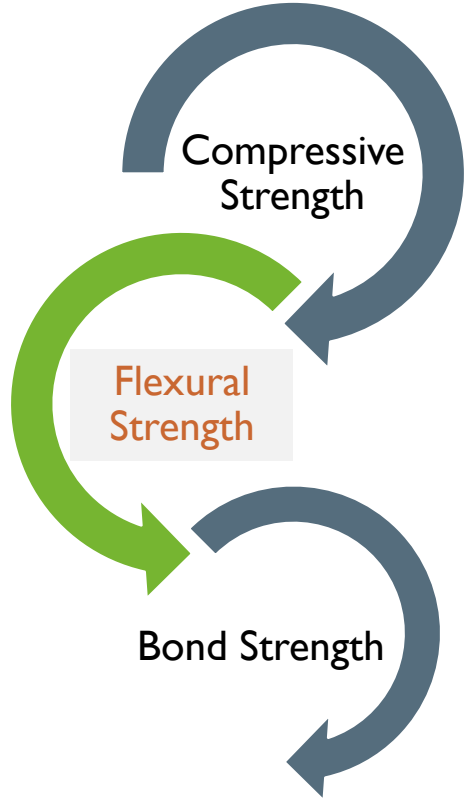
Mechanical Properties

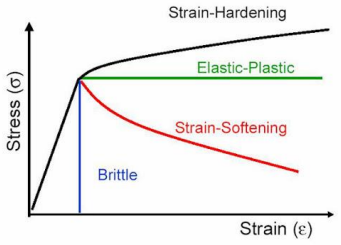
- ASTM C348.
- Flexural Strength = $\frac{3PL}{2bd^2}$ [38].



Compression Test Setup [39]

P: failure force (N)
L: effective span between supports (mm)
b: width (mm)
d: depth (mm)



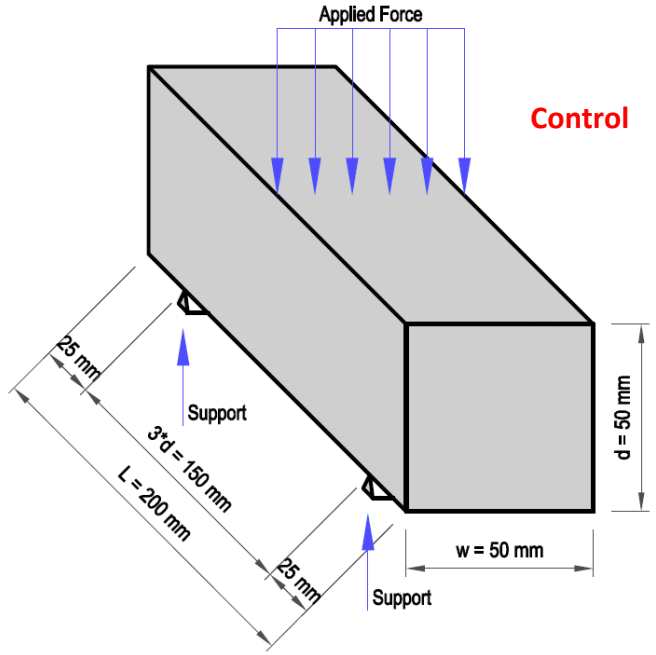


Methodology

Mechanical Properties

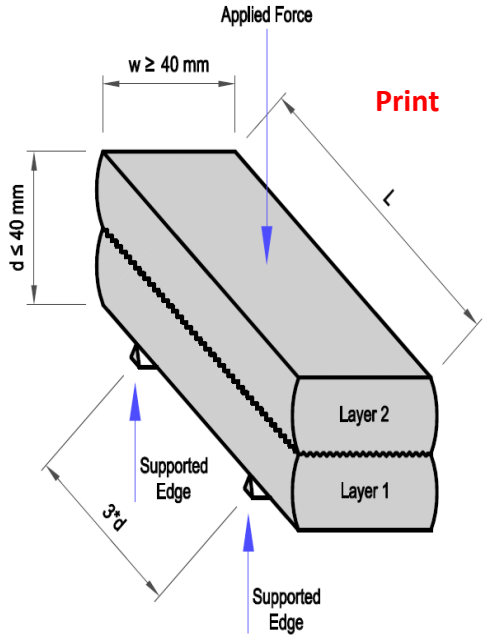
Specimens

Control

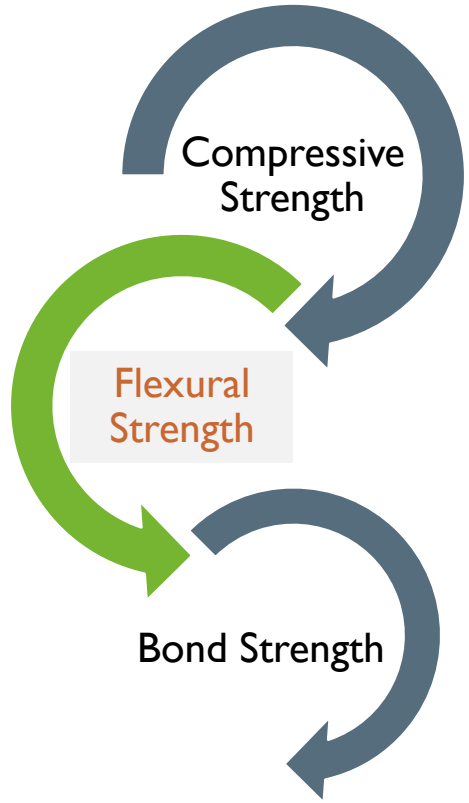


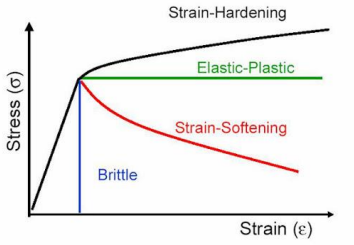
Ambient

Print



Ambient Site





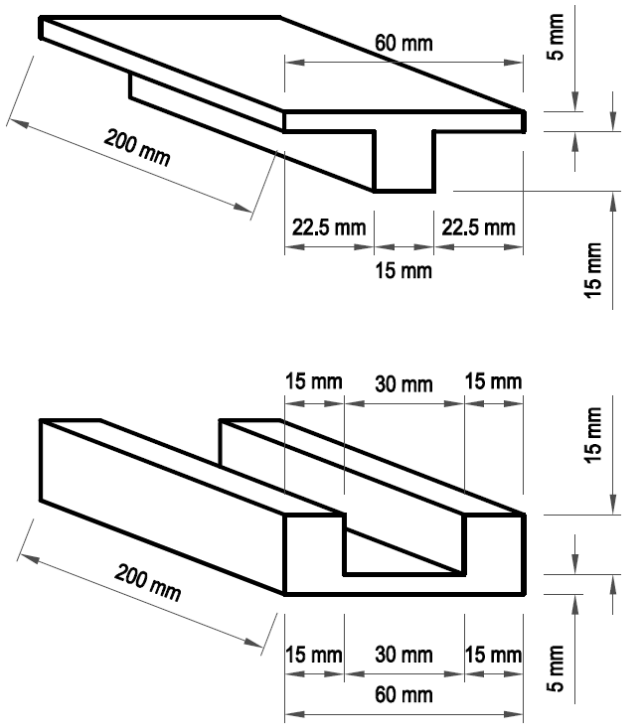
Methodology

Mechanical Properties

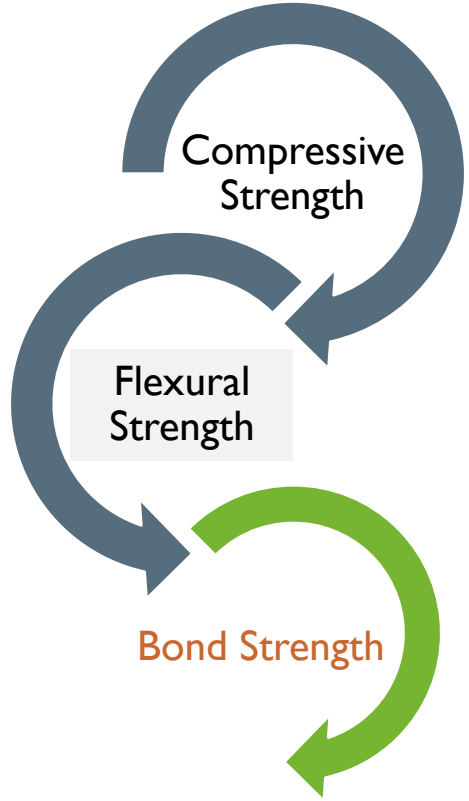
- A simplified novel setup was designed for bond test.

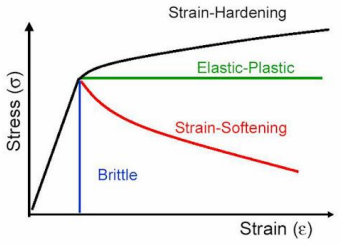


Bond Shear Test Setup



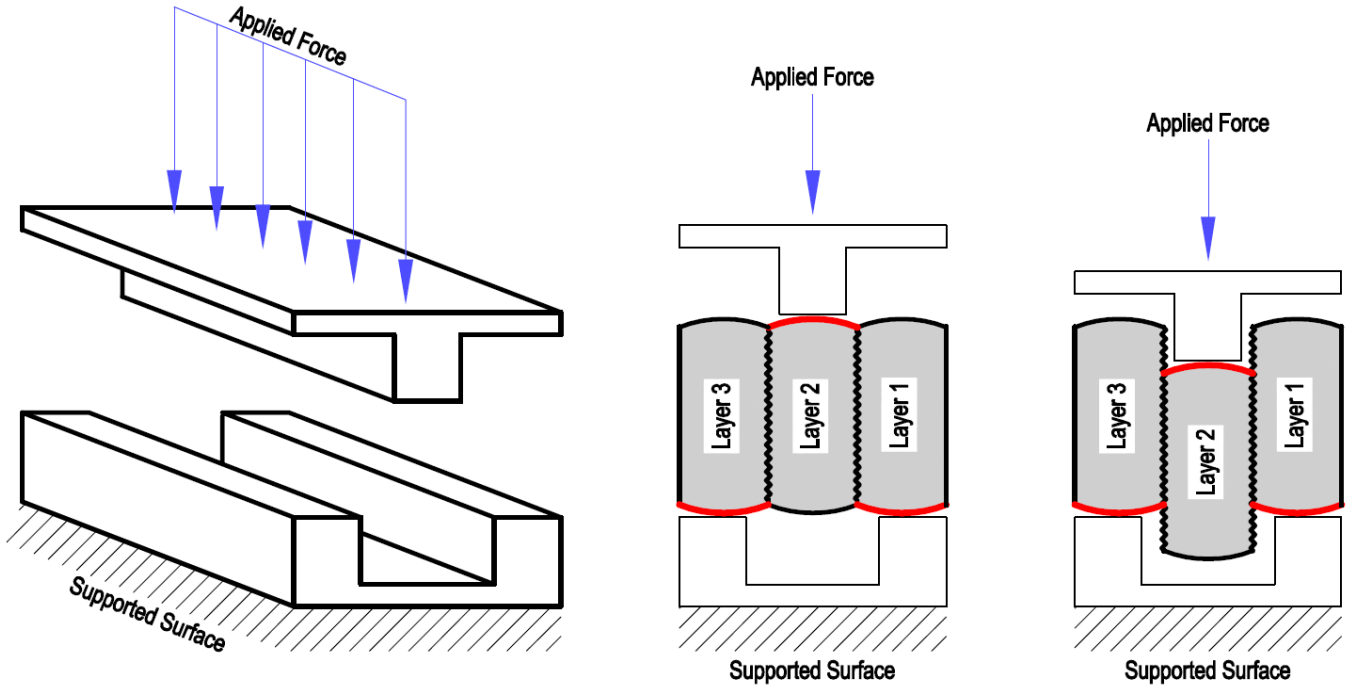
Sketch of Bond Shear Test Fixture



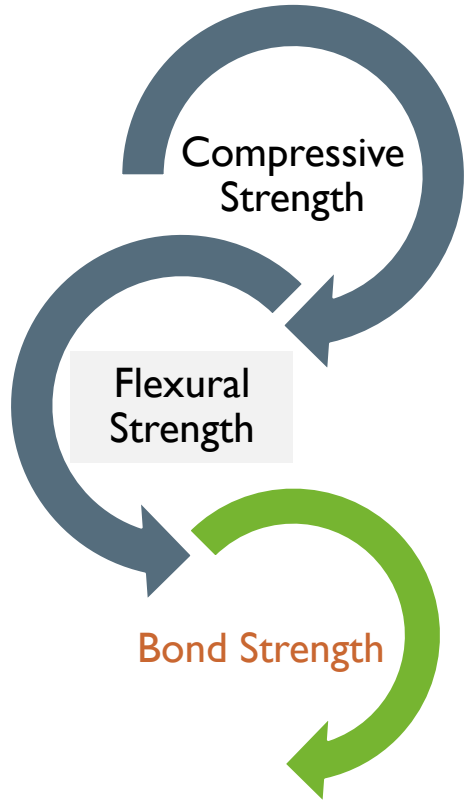


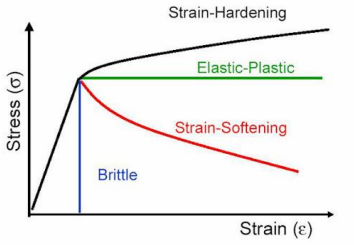
Methodology

Mechanical Properties



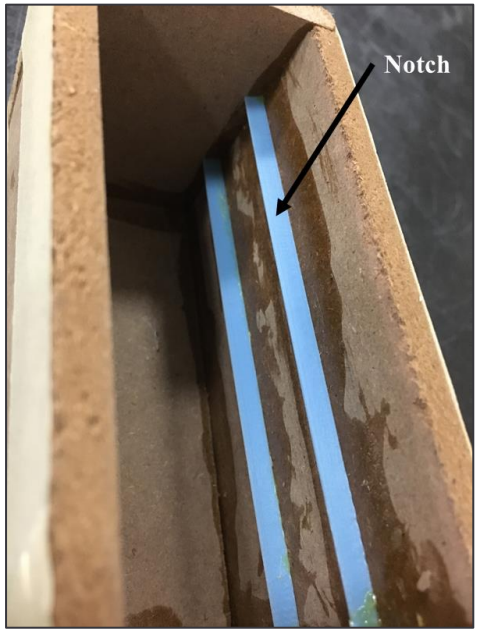
Bond Shear Test Load Mechanism

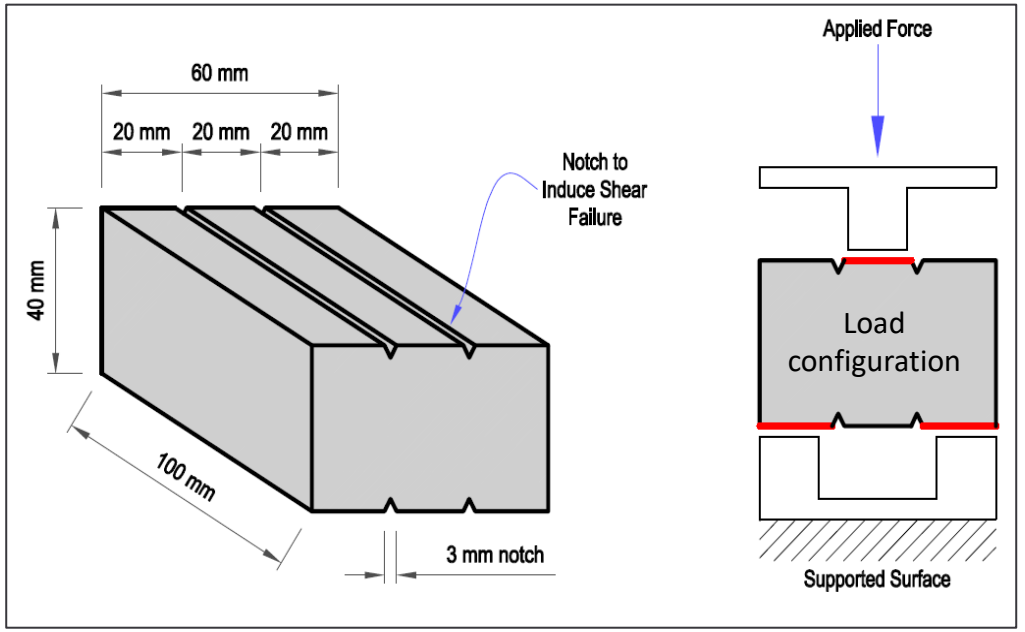




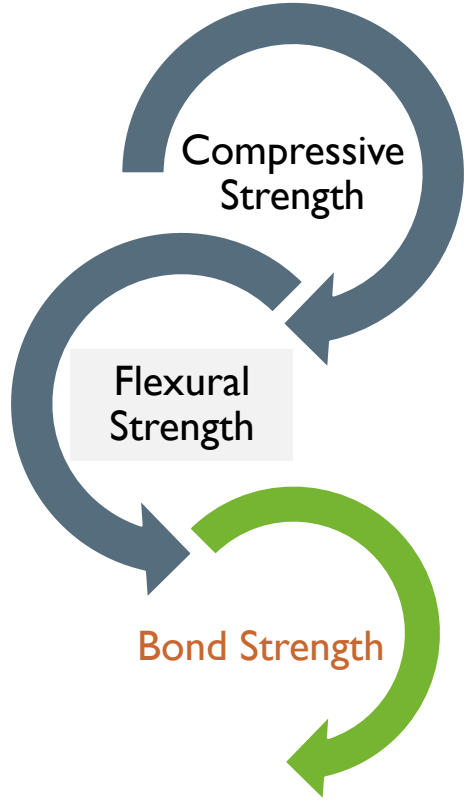
Methodology

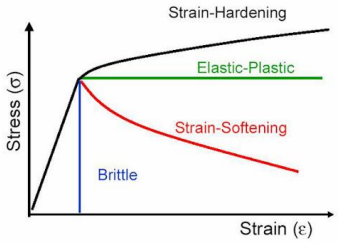
Mechanical Properties





Bond Shear **Control** Specimen Mold and Load Configuration





Methodology

Mechanical Properties

Specimens

Control

Ambient

t = 30 s

t = 4 hrs

t = 2.5 hrs

Print

Ambient

t = 30 s

t = 4 hrs

t = 2.5 hrs

Site

t = 30 s

t = 4 hrs

t = 2.5 hrs

Compressive Strength

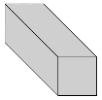
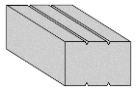
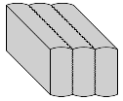
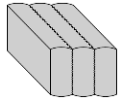
Flexural Strength

Bond Strength

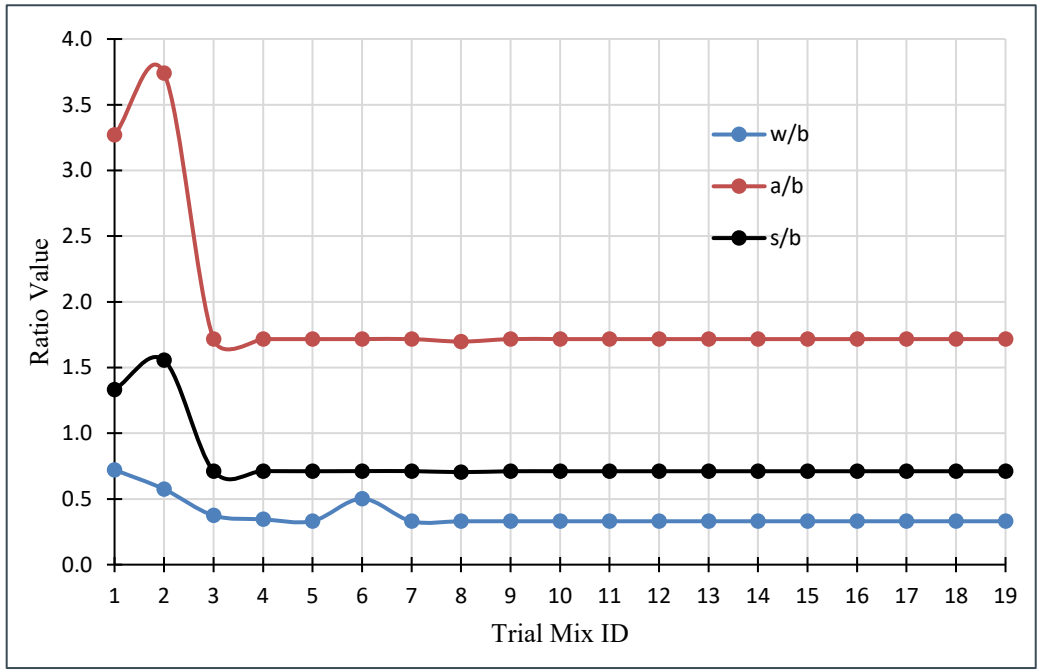
Experimental Program

Description	Parameter	Experimental Scenarios	
		Scenario 1	**Scenario 2**
General details for testing scenarios	Mixing water condition	Cool water from the laboratory's faucet	Water heated up to 50˚ C
	Curing	7 days for control (molded) specimens only	
		No curing for all printed specimens	
	Storage prior to testing	Molded specimens are stored in the curing tank inside the laboratory for 7 day	Molded specimens are stored in the curing tank outside the laboratory for 7 day
		Printed specimens are stored without curing inside the laboratory for 7 days	Printed specimens are stored without curing outside the laboratory for 7 days
Tests for fresh concrete properties	Flow table	3 repetitions	3 repetitions
	Open time	3 repetitions	3 repetitions
	Extrudability	3 repetitions	3 repetitions
Shape Stability		Printed specimens for flexural and bond shear tests will be used for shape stability evaluation	Printed specimens for flexural and bond shear tests will be used for shape stability evaluation

Experimental Program

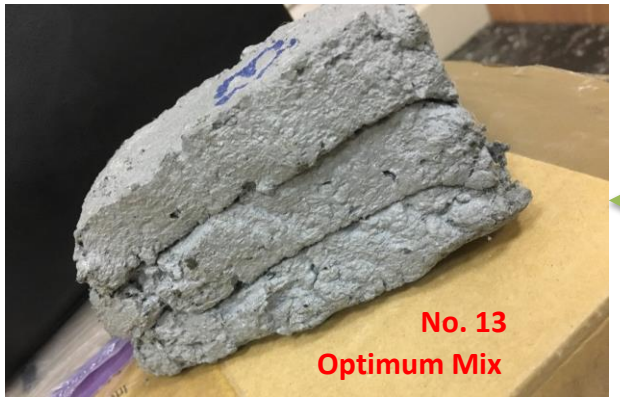
Tests for hardened concrete properties	Compressive strength	 3 control specimens: CONTROL-CT-01 CONTROL-CT-02 CONTROL-CT-03	 6 printed specimens: AK-01-CT-01 AK-01-CT-02 AK-01-CT-03 AK-01-CT-04 AK-01-CT-05 AK-01-CT-06	 6 printed specimens: AK-02-CT-01 AK-02-CT-02 AK-02-CT-03 AK-02-CT-04 AK-02-CT-05 AK-02-CT-06
	Flexural strength	 3 control specimens: CONTROL-FT-01 CONTROL-FT-02 CONTROL-FT-03	 4 printed specimens: AK-01-FT-01 AK-01-FT-02 AK-01-FT-03 AK-01-FT-04	 4 printed specimens: AK-02-FT-01 AK-02-FT-02 AK-02-FT-03 AK-02-FT-04
	Bond shear strength	 9 control specimens: 1. Time interval = 30 s CONTROL-BT-30S-01 CONTROL-BT-30S-02 CONTROL-BT-30S-03 2. Time interval = 2.5 hours CONTROL-BT-2.5H-01 CONTROL-BT-2.5H-02 CONTROL-BT-2.5H-03 3. Time interval = 4 hours CONTROL-BT-4H-01 CONTROL-BT-4H-02 CONTROL-BT-4H-03	 9 specimens: 1. Time interval = 30 s AK-01-BT-30S-01 AK-01-BT-30S-02 AK-01-BT-30S-03 2. Time interval = 2.5 hours AK-01-BT-2.5H-01 AK-01-BT-2.5H-02 AK-01-BT-2.5H-03 3. Time interval = 4 hours AK-01-BT-4H-01 AK-01-BT-4H-02 AK-01-BT-4H-03	 9 specimens: 1. Time interval = 30 s AK-02-BT-30S-01 AK-02-BT-30S-02 AK-02-BT-30S-03 2. Time interval = 2.5 hours AK-02-BT-2.5H-01 AK-02-BT-2.5H-02 AK-02-BT-2.5H-03 3. Time interval = 4 hours AK-02-BT-4H-01 AK-02-BT-4H-02 AK-02-BT-4H-03

Trial Batches



w/b, a/b, and s/b Variations Among Trial Mixes.





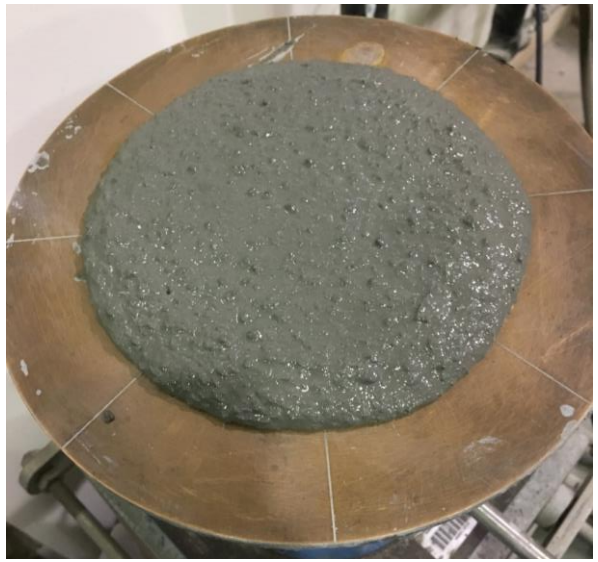
Type	Tensile Strength	Modulus of Elasticity	Fiber Length	Fiber Diameter	Specific Gravity
Ultra-High Modulus Polyethylene Fiber	2500 MPa	70 GPa	12 mm	38 μm	0.96

Fresh Properties

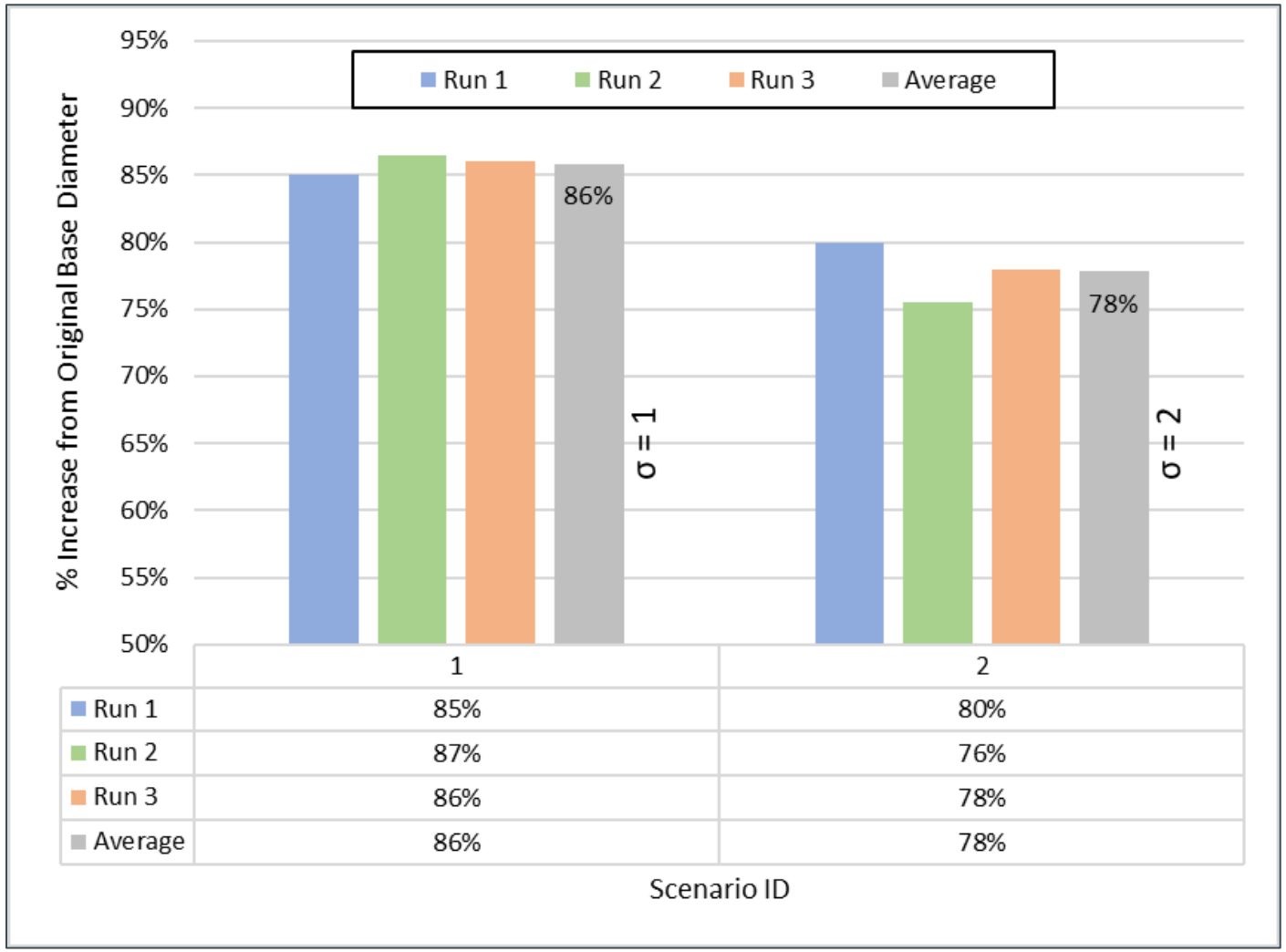


Flow Table

- The drop in flow between both scenarios is 9%.
- Flow in the optimum mix is very sensitive to short time intervals.



Flow Table Test



Flow Table Test Results for Both Scenarios



Fresh Properties

Open Time & Extrudability

- For Scenario 1 and 2, Open time was found to be 7.2 and 6 minutes respectively. (drop = 16%)
- Loss of workability in the second scenario due to the high temperature of the mix.
- Short open times are suitable for sequential batching applications.

- Mix was extrudable using normal nozzle 20 x 40 $mm^{2.}$
- Mix was not extrudable using smaller nozzle 10 x 40 $mm^{2.}$
- Extrudability is related to open time. A mix can still be extrudable beyond the open time of a mix, but with higher pressure and tendency to segregation or clogging.



Stoppage of Printing



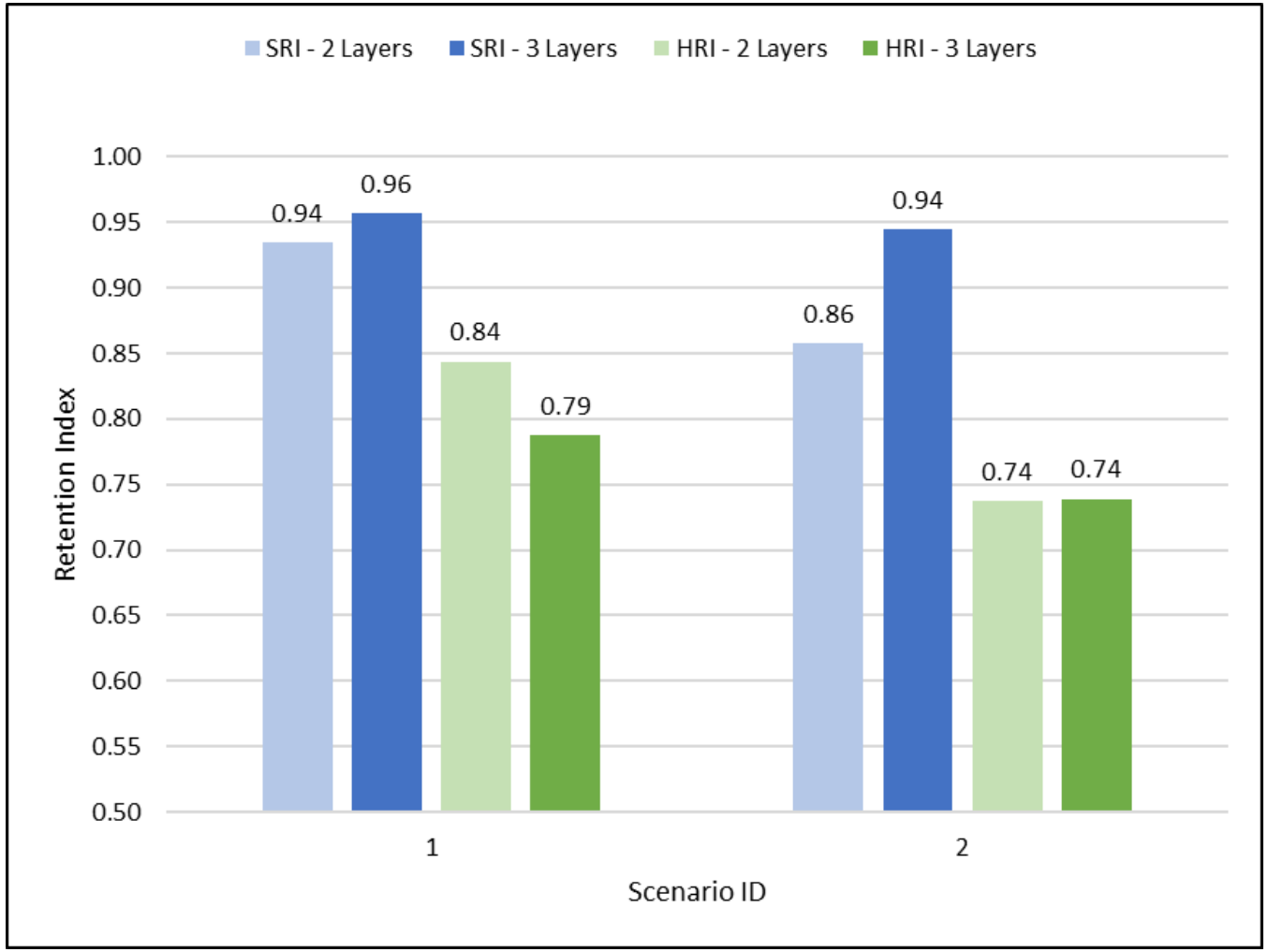
Scenario 1



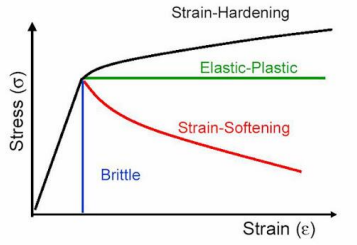
Scenario 2

Shape Retention

- Drop from S1 to S2:
  - 8.5% for 2-layers SRI
  - 2% for 3-layers SRI
  - 12% for 2-layers HRI
  - 6% for 3-layers HRI
- The use of long filaments give a more proper insight on shape stability.
- Higher **early-age** flowability in site conditions.
- The values of SRI fell in recommended range by Zareiyan and Khoshnevis [22].



Shape Retention Indices for Both Scenarios

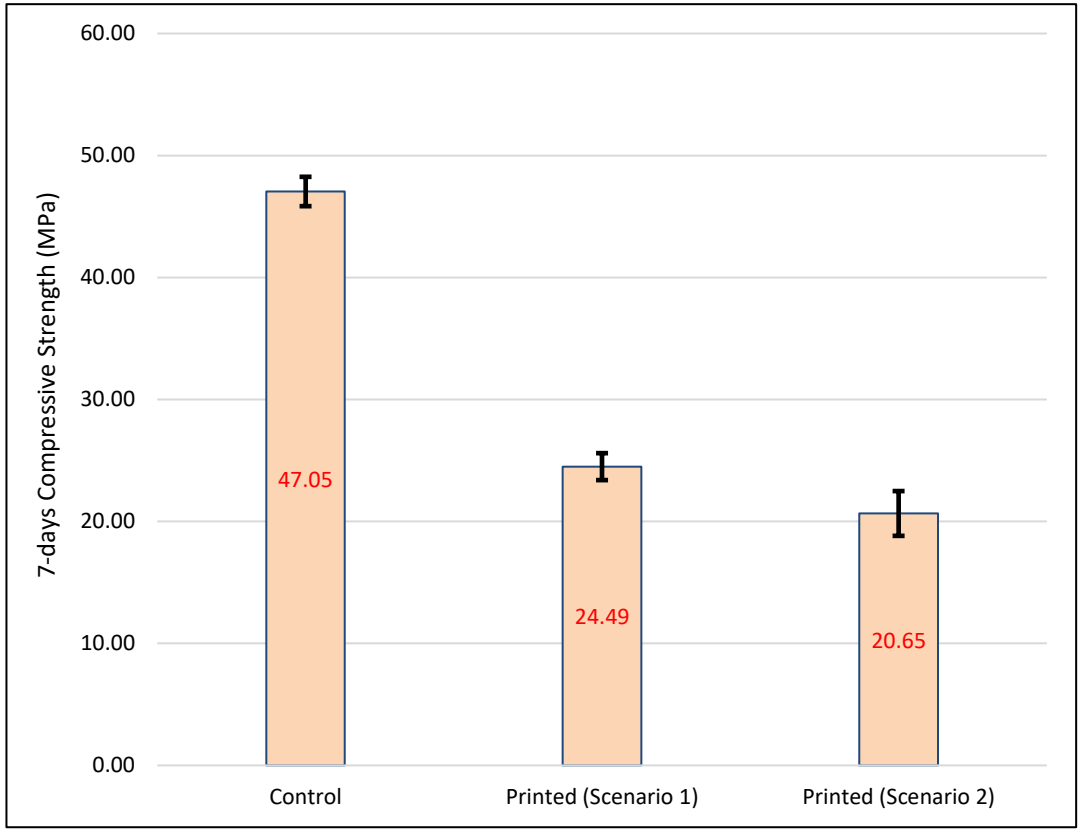


Mechanical Properties

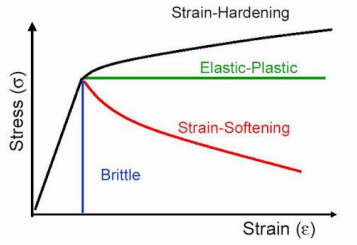
Compressive Strength

Results

- Compressive strength dropped from control to ambient print, and site print.



Summary of Compressive Strength Results of Control Cubes and Printed Cubes in both Scenarios.



Mechanical properties

Flexural Strength

Failure Pattern

- All printed prisms failed by tensile stresses rather than interface shear fracture.
- Bond is maintained.
- Fiber orientation parallel to printing direction is significant to the flexural strength. Bridging effect is evident.



Side View

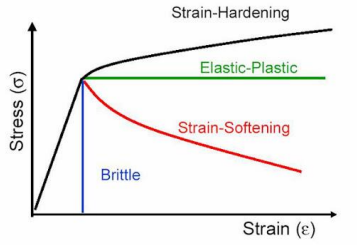


Bottom View





Section



Mechanical Properties

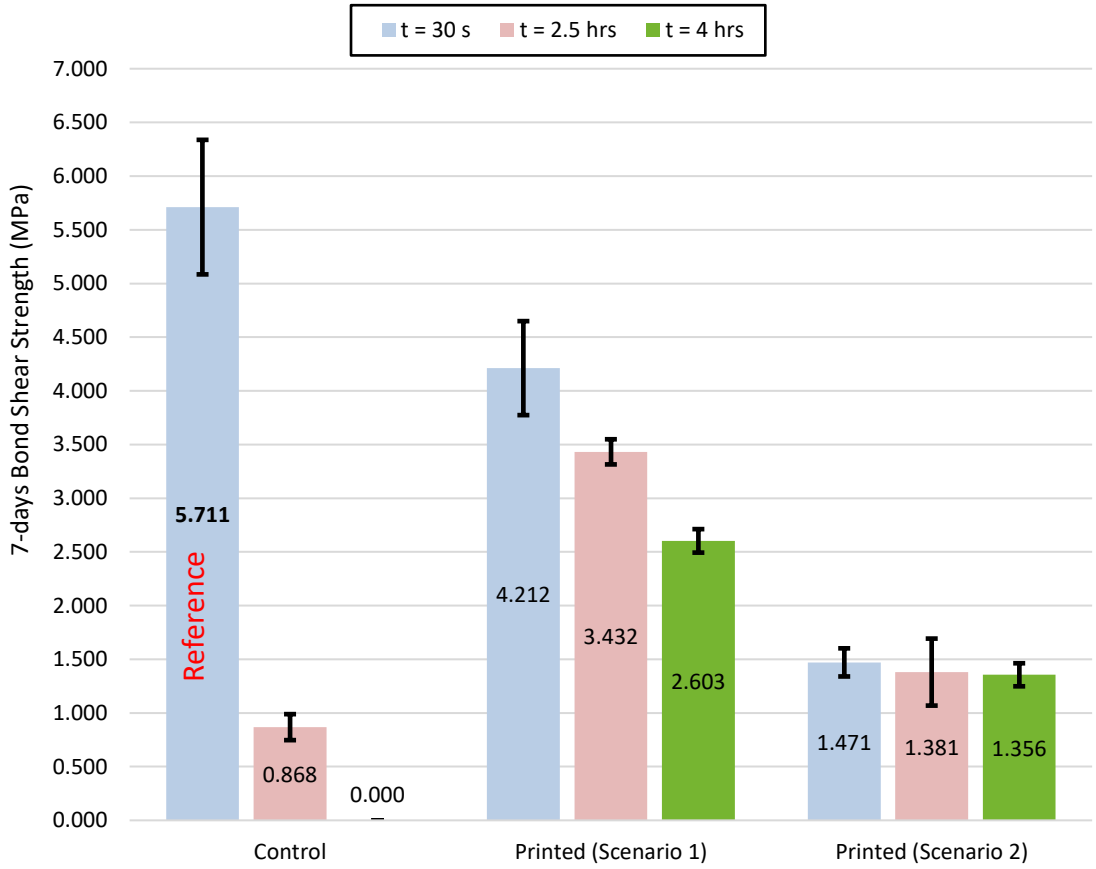
Bond Shear Strength

Results

- Bond Shear Strength (BSS) was found to decrease with time interval (gap) between layers.
- Trend of reduction is more dramatic in molded specimens.
- Achieved values were relatively good compared to literature, that is by Rahul, Santhanam, Meena, and Ghani [27].
- Silica fume improves packing of the layered material [38].



Bond Test Setup



Summary of Bond Shear Strength Results

Two more New Innovations are under Consideration: 2026 – 2027

2. MSc.:

OPTIMIZING FIBER ALIGNMENT IN 3D CONCRETE PRINTING: DEVELOPMENT OF A NOVEL NOZZLE HEAD FOR ENHANCED REINFORCEMENT EFFICIENCY.

3. PhD.:

PATHWAYS TO CARBON NEUTRALITY: A DECISION SUPPORT TOOL FOR OPTIMIZING CARBON SEQUESTRATION OF DESERT PLANTS-A SHARJAH CASE STUDY.

THANK YOU
Questions?